# INVERTIS UNIVERSITY, BAREILLY

**Department of Computer Applications — Project-Based Learning (PBL)**

## PROJECT REPORT: Food Delivery Time and Order Analysis

| PROJECT ID | GROUP ID | PROGRAM & SEM | PROJECT MENTOR |
|---|---|---|---|
| DA-005 | BCA_AI_05 | BCA AI (Sem I, Sec A) | Mr. Vishnu |

## 1. EXECUTIVE SUMMARY & ABSTRACT

This project presents an end-to-end data analytics study on a real-world food delivery dataset comprising **1,000 order records** (`Food_Delivery_Times.csv`). The primary objective is to evaluate operational bottlenecks causing delivery delays beyond the standard **45-minute Service Level Agreement (SLA) threshold**. Using Python, Pandas, Matplotlib, and Seaborn, the project performs data cleaning, mode/median imputation for missing records, feature engineering, quantitative KPI calculation, and visual exploratory analysis.

## 2. PROBLEM STATEMENT & OBJECTIVES

Inaccurate Estimated Time of Arrival (ETA) predictions leading to late food deliveries (>45 minutes) diminish customer retention and disrupt logistics. The project aims to:

- Clean raw operational records by imputing missing values across weather, traffic, and driver experience.
- Compute core descriptive metrics (Mean, Median, Min, Max durations, and Delay proportion).
- Identify delay correlations across transit distance, time of day, and traffic conditions.
- Develop 4 visual analytics charts and provide data-backed operational solutions.

## 3. ANALYTICAL KPIS & PERFORMANCE SUMMARY

| Metric Name | Calculated Value | Operational Insight & SLA Context |
|---|---|---|
| **Total Sample Analyzed** | 1,000 Orders | Complete dataset records evaluated. |
| **Average Delivery Time** | 56.73 minutes | Overall mean duration across all orders. |
| **Median Delivery Time** | 55.50 minutes | Midpoint transit duration. |
| **SLA Delay Benchmark** | 45.0 minutes | Standard maximum on-time threshold limit. |
| **Total Delayed Orders** | 664 (66.40%) | Orders exceeding the 45-minute SLA limit. |
| **Min / Max Duration** | 8.0m / 153.0m | Fastest completed order vs maximum delayed outlier. |

## 4. KEY ANALYTICAL FINDINGS ACROSS VISUALIZATIONS

- **Distance Threshold Impact:** Orders exceeding **10 km** register a >90% delay probability, proving distance is the primary operational bottleneck.
- **Peak-Hour Congestion:** Morning and Evening slots exhibit the highest total order load and account for over 72% of total delay instances.
- **Traffic Severity:** High Traffic conditions push median delivery duration up to **~65 minutes**.

## 5. STRATEGIC BUSINESS RECOMMENDATIONS

- **Dynamic ETA Buffering:** Inject a 15–20 minute buffer to estimated customer ETAs during peak traffic hours and adverse weather conditions.
- **Geofenced Distance Caps:** Restrict maximum delivery radius to 8 km during peak lunch (11 AM–2 PM) and dinner (7 PM–9 PM) rush windows.
- **Experienced Courier Dispatch:** Route couriers with >3 years of experience to high-distance (>10 km) or high-traffic orders.

## 6. GROUP MEMBER ROLES & TASK ALLOCATION (GROUP BCA_AI_05)

| Sr. | Student Name | Student ID | Roll No. | Project Role & Contribution Breakdown |
|---|---|---|---|---|
| **1** | **SUDHEER KUMAR** | **BCAI2026069** | **2610206191** | **Project Lead & Technical Head (Complete Python Coding, Data Preprocessing, Imputation, Chart Generation & GitHub Deployment)** |
| 2 | VYOM SHARMA | BCAI2026071 | 2610206218 | Presentation Team (Problem Statement & Introduction) |
| 3 | SUBRAT MISHRA | BCAI2026082 | 2610206190 | Presentation Team (KPI Dashboard & Summary Metrics) |
| 4 | ASHISH KUMAR | BCAI2026131 | 2610206046 | Presentation Team (Delivery Time Distribution Chart) |
| 5 | ADITY PRATAP SINGH | BCAI2026136 | 2610206010 | Presentation Team (Distance vs Duration Scatter Plot) |
| 6 | FAZILAT KHAN | BCAI2026202 | 2610206072 | Presentation Team (Time of Day Volume & Peak Hours) |
| 7 | SAMEER RAJ | BCAI2026223 | 2610206161 | Presentation Team (Traffic Impact & Box Plot Analysis) |
| 8 | SHALINI KUMARI | BCAI2026232 | 2610206173 | Presentation Team (Business Recommendations) |
| 9 | RAHUL KUMAR | BCAI2026235 | 2610206147 | Presentation Team (Tech Stack & Repository Overview) |
| 10 | HIMANSHU KUMAR CHANDEL | BCAI2026236 | 2610206079 | Presentation Team (Conclusion & Q&A Coordination) |